

Operators and Expressions

**SIMATS**
ENGINEERING

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C PROGRAMMING

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QUESTIONS 8 / 8 completed

Swap 2 No's

SWAP 2 NO'S

C Program to Swap Two Numbers.
Swapping two numbers in C programming means swapping the values of two variables.
For example, there are two variables m & n. Value of m is "2" & value of n is "3".
Before Swapping: m ...

PROGRAM EDITOR C Run Save

INPUT

2
3

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QUESTIONS 8 / 8 completed

Arithmetic

ARITHMETIC

Write a program that takes two numbers as input and prints their sum, difference, product, and quotient.

PROGRAM EDITOR C Run Save

INPUT

4
2

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Convert

CONVERT

Write a program that converts Celsius to Fahrenheit. The formula is Fahrenheit = (Celsius * 1.8) + 32.

PROGRAM EDITOR C Run Save

INPUT

2



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Rectangle

RECTANGLE
Write a program that calculates the area of a rectangle given its length and width.
Problem Description: Write a program that calculates the area

PROGRAM EDITOR
C Run Save

```
1 #include <stdio.h>
2 int main(){
3     int l,w,b;
4     printf("Enter the Length of the Rectangle: ");
5     scanf("%d", &l);
6     printf("Enter the Width of the Rectangle: ");
7     scanf("%d", &w);
8     b= l*w;
9     printf("The area of the Rectangle is: %d",b);
10    return 0;
11 }
```

INPUT
2
4



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QUESTIONS 8 / 8 completed

ODD EVEN

ODD EVEN
Write a program that checks whether a number is even or odd.

PROGRAM EDITOR
C Run Save

```
1 #include <stdio.h>
2 int main(){
3     int a;
4     printf("Enter the number: ");
5     scanf("%d", &a);
6     if(a%2==0){
7         printf("The Given number is even");
8     }
9     else{
10        printf("The Given number is odd");
11    }
12 }
```

INPUT
1



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QUESTIONS 8 / 8 completed

Leap

LEAP
Write a program that determines whether a year is a leap year or not. A year is a leap year if it is divisible by 4, but not divisible by 100 unless it is also divisible by 400.

PROGRAM EDITOR
C Run Save

```
1 #include <stdio.h>
2 int main(){
3     int year;
4     printf("Enter the year to find out Leap year or not: \n");
5     scanf("%d",&year);
6     if(year%4==0 && year%100!=0 || year%400==0){
7         printf("The given Year is Leap year: %d\n",year);
8     }
9     else{
10        printf("The given Year is not Leap year: %d\n",year);
11    }
12 }
```

INPUT
2023



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QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN
Write a program that takes three integers as input and prints the maximum and minimum of the three.

PROGRAM EDITOR
C Run Save

```
1 #include <stdio.h>
2 int main(){
3     int a,b,c,max,min;
4     printf("Enter the three values: \n");
5     scanf("%d %d %d",&a,&b,&c);
6     max=(a>b)?((a>c)?a:c):((b>c)?b:c);
7     min=(a<b)?((a<c)?a:c):((b<c)?b:c);
8     printf("The Maximum value of the three integers: %d\n",max);
9     printf("The Minimum value of the three integers: %d\n",min);
10    return 0;
11 }
```

INPUT
1
2
3

QUESTIONS 8 / 8 completed

Sum-Avg

SUM-AVG

Write a program that takes a list of numbers as input and computes the sum and average of the numbers.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main(){
3     int n,i,sum=0, a[10];
4     float avg;
5     printf("Enter the number of elements: \n");
6     scanf("%d",&n);
7     printf("Enter the %d number of elements: \n");
8     for(i=0;i<n;i++){
9         scanf("%d",&a[i]);
10        sum+=a[i];
11    }
12    avg=(float) sum/n;
13    printf("Sum of the values: %d\n",sum);
14    printf("Average of the values: %.2f\n",avg);
15    return 0;
16 }
```

INPUT

```
5
10 20 30 40 50
```

Decision Macking

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QUESTIONS 8 / 8 completed

+ OR -

+ OR -

C program to Check Whether a Number is Positive or Negative.

Problem Description
The program takes the given integer and checks whether the integer is positive or negative.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 int main(){
3     int a;
4     printf("Enter the number to check if its positive or negative: \n");
5     scanf("%d",&a);
6     if(a>0){
7         printf("The given number is Positive: %d\n",a);
8     }
9     else if(a<0){
10        printf("The given number is Negative: %d\n",a);
11    }
12    else{
13        printf("The given number is Zero: %d\n",a);
14    }
15 }
```

INPUT

0

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QUESTIONS 8 / 8 completed

Greatest

GREATEST

Write a program that takes two numbers as input and determines which one is larger.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 int main(){
3     int a;
4     printf("Enter the number to check if its positive or negative: \n");
5     scanf("%d",&a);
6     if(a>0){
7         printf("The given number is Positive: %d\n",a);
8     }
9     else if(a<0){
10        printf("The given number is Negative: %d\n",a);
11    }
12    else{
13        printf("The given number is Zero: %d\n",a);
14    }
15 }
```

INPUT

0

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QUESTIONS 8 / 8 completed

Character

CHARACTER

Write a program that takes a character as input and determines whether it is a vowel or a consonant.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 #include <ctype.h>
3 int main(){
4     char ch;
5     printf("Enter Character: \n");
6     scanf("%c", &ch);
7     ch=tolower(ch);
8     if(ch >= 'a' && ch <= 'z'){
9         if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u'){
10             printf("%c is a vowel.\n",ch);
11         }
12         else{
13             printf("%c is a constant.\n",ch);
14         }
15     }
16     else{
17         printf("Invalid input!");
18     }
19     return 0;
20 }
```

INPUT

A



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QUESTIONS 8 / 8 completed

Age

AGE

Write a program that takes an age as input and determines whether the person is a child, teenager, adult, or senior citizen.

PROGRAM EDITOR

C

Run Save

```

1 #include <stdio.h>
2 int main(){
3     int age;
4     printf("Enter the age:\n");
5     scanf("%d",&age);
6     if(age<0){
7         printf("Invalid age enter the correct age\n");
8     }
9     else if(age<=12){
10        printf("Child\n");
11    }
12    else if(age>=13 && age<=19){
13        printf("Teenager\n");
14    }
15    else if(age>=20 && age<=59){
16        printf("Adult\n");
17    }
18    else{
19        printf("Senior citizen\n");
20    }
21    return 0;

```

INPUT

15



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QUESTIONS 8 / 8 completed

Temperature

TEMPERATURE

Write a program that takes a temperature as input and determines whether it is hot, warm, or cold.

PROGRAM EDITOR

C

Run Save

```

1 #include<stdio.h>
2 int main(){
3     int temp;
4     printf("Enter the Temperature:\n");
5     scanf("%d",&temp);
6     if(temp>30){
7         printf("Hot\n");
8     }
9     else if(temp>=20 && temp<=30){
10        printf("Warm\n");
11    }
12    else{
13        printf("Cold");
14    }
15    return 0;
16 }

```

INPUT

25



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QUESTIONS 8 / 8 completed

Ups-Lows

UPS-LOWS

Write a program that takes a character as input and determines whether it is uppercase or lowercase.

PROGRAM EDITOR

C

Run Save

```

1 #include <stdio.h>
2 int main(){
3     char ch;
4     printf("Enter the Letter:\n");
5     scanf("%c", &ch);
6     if(ch>='A' && ch<='Z'){
7         printf("The given Letter is Uppercase\n");
8     }
9     else if(ch>='a' && ch<='z'){
10        printf("The given Letter is Lowercase\n");
11    }
12    else{
13        printf("Not a Letter\n");
14    }
15    return 0;
16 }

```

INPUT

A



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QUESTIONS 8 / 8 completed

Multiple

MULTIPLE

Write a program that takes a number as input and determines whether it is a multiple of 3 or 5.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 int main(){
3     int a;
4     printf("Enter the number:\n");
5     scanf("%d",&a);
6     if(a%3==0 && a%5==0){
7         printf("The given number is a multiple of both 3 and 5\n");
8     }
9     else if(a%5==0){
10        printf("The given number is a multiple of 5\n");
11    }
12    else if(a%3==0){
13        printf("The given number is a multiple of 3\n");
14    }
15    else{
16        printf("The given number is not a multiple of both 3 and 5\n");
17    }
18    return 0;
19 }
```

INPUT

30



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QUESTIONS 8 / 8 completed

Grade

GRADE

Write a program that takes a grade as input and determines the corresponding letter grade (A, B, C, D, or F).

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 int main(){
3     int score;
4     printf("Enter the Score:\n");
5     scanf("%d",&score);
6     if(score<0 || score>100){
7         printf("Invalid Input\n");
8         return 0;
9     }
10    switch(score/10){
11        case 10: case 9:
12            printf("A Grade\n");
13            break;
14        case 8:
15            printf("B Grade\n");
16            break;
17        case 7:
18            printf("C Grade\n");
19            break;
20        case 6:
```

INPUT

10

```
21     printf("D Grade\n");
22     break;
23     default:
24         printf("F Grade\n");
25         break;
26 }
27 return 0;
28 }
```

Looping

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QUESTIONS 18 / 18 completed

Sum(ODD,EVEN)

SUM(ODD,EVEN)

C Program to Find the Sum of Even and Odd Numbers.
Problem Description
The program takes the number N and finds the sum of odd and even numbers from 1 to N.

Testcases:
-5
10
*

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 int main(){
3     int i,n;
4     int sum_even=0,sum_odd=0;
5     printf("Enter the number:\n");
6     scanf("%d",&n);
7     for(i=1;i<=n;i++){
8         if(i%2==0){
9             sum_even+=i;
10        }
11        else{
12            sum_odd+=i;
13        }
14    }
15    printf("Sum of even numbers:%d\n",sum_even);
16    printf("Sum of odd numbers:%d\n",sum_odd);
17    return 0;
18 }
```

INPUT

10

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QUESTIONS 18 / 18 completed

Natural

NATURAL

Write a program that prints the first n natural numbers using a for loop.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 int main(){
3     int n,i;
4     printf("Enter the number:\n");
5     scanf("%d",&n);
6     for(i=1;i<=n;i++){
7         printf(" %d",i);
8     }
9     printf("\n");
10    return 0;
11 }
```

INPUT

6

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QUESTIONS 18 / 18 completed

Sum

SUM

Write a program that calculates the sum of the first n natural numbers using a while loop.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 int main(){
3     int n,i=1,sum=0;
4     printf("Enter the number:\n");
5     scanf("%d",&n);
6     while(i<=n){
7         sum+=i;
8         i++;
9     }
10    printf("Sum = %d\n",sum);
11    return 0;
12 }
```

INPUT

10



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QUESTIONS 18 / 18 completed

Table

TABLE

Write a program that prints the multiplication table of a given number using a for loop.

PROGRAM EDITOR

C

Run

Save

```

1 #include <stdio.h>
2 int main(){
3     int i,n;
4     printf("Enter the number for Multiplication table:\n");
5     scanf("%d",&n);
6     for(i=1;i<=10;i++){
7         printf("%d x %d = %d\n",n,i,n*i);
8     }
9     return 0;
10 }

```

INPUT

Your INPUT goes here!



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QUESTIONS 18 / 18 completed

Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

PROGRAM EDITOR

C

Run

Save

```

1 #include <stdio.h>
2 int main(){
3     int n,i=2;
4     printf("Enter the number:\n");
5     scanf("%d",&n);
6     if(n<=1){
7         printf("Not Prime\n");
8         return 0;
9     }
10    while(i<n){
11        if(n%i==0){
12            printf("Not Prime\n");
13            return 0;
14        }
15        i++;
16    }
17    printf("Prime\n");
18    return 0;
19 }

```

INPUT

4



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QUESTIONS 18 / 18 completed

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

PROGRAM EDITOR

C

Run

Save

```

1 #include <stdio.h>
2 int main(){
3     int count=0;
4     char str[100];
5     printf("Enter the string:\n");
6     scanf("%s",str);
7     for(int i=0;str[i]!='\0';i++){
8         if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' || str[i] == 'o' || str[i] == 'u' ||
9             str[i] == 'A' || str[i] == 'E' || str[i] == 'I' || str[i] == 'O' || str[i] == 'U')
10            count++;
11    }
12    printf("%d\n",count);
13    return 0;
14 }

```

INPUT

hello



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QUESTIONS 18 / 18 completed

Fibonacci

FIBONACCI

Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR

C

Run

Save

```

1 #include <stdio.h>
2 int main(){
3     int first=0,second=1,next,limit;
4     printf("Enter the number for limit of Fibonacci series:\n");
5     scanf("%d", &limit);
6     while(first<=limit){
7         printf(" %d",first);
8         next= first+second;
9         first = second;
10        second= next;
11    }
12    printf(" \n");
13    return 0;
14 }

```

INPUT

7



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QUESTIONS 18 / 18 completed

Factorial

FACTORIAL

Write a program that takes a number as input and prints its factorial using a for loop.

PROGRAM EDITOR

C

Run

Save

```

1 #include <stdio.h>
2 int main(){
3     int n;
4     long long factorial =1;
5     printf("Enter the factorial number:\n");
6     scanf("%d",&n);
7     for(int i=1;i<=n;i++){
8         factorial*=i;
9     }
10    printf("%lld\n",factorial);
11    return 0;
12 }

```

INPUT

5



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QUESTIONS 18 / 18 completed

Pyramid

PYRAMID

Write a program that prints a pyramid pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

```

1 #include <stdio.h>
2 int main(){
3     int n;
4     printf("Enter the number:\n");
5     scanf("%d",&n);
6     for(int i=1;i<=n;i++){
7         for(int j=1;j<=n-i;j++){
8             printf(" ");
9         }
10        for(int k=1;k<=(2*i-1);k++){
11            printf("*");
12        }
13        printf("\n");
14    }
15    return 0;
16 }

```

INPUT

7

QUESTIONS 18 / 18 completed

Diamond

DIAMOND

Write a program that prints a diamond pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

INPUT

10

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     printf("Enter the number:\n");
5     scanf("%d",&n);
6     for(int i=1; i<=n;i++){
7         for(int j=1;j<=n-i;j++){
8             printf(" ");
9         }
10        for(int k=1;k<=(2*i-1);k++){
11            printf("*");
12        }
13        printf("\n");
14    }
15    for(int i=n-1;i>=1;i--){
16        for(int j=1;j<=n-i;j++){
17            printf(" ");
18        }
19        for(int k=1;k<=(2*i-1);k++){
20            printf("*");
```

```
21    }
22    printf("\n");
23 }
24     return 0;
25 }
```

QUESTIONS 18 / 18 completed

R- triangle

R- TRIANGLE

Write a program that prints a right-angled triangle pattern of stars using nested loops.

PROGRAM EDITOR

C

Run

Save

INPUT

5

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     for(int i=1;i<=n;i++){
6         for(int j=1;j<=i;j++){
7             printf("*");
8         }
9         printf("\n");
10    }
11    return 0;
12 }
```

QUESTIONS 18 / 18 completed

Hollow Square ▾

HOLLOW SQUARE

Write a program that prints a hollow square pattern of stars using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

INPUT

7

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     for(int i=1;i<=n;i++){
6         for(int j=1;j<=n;j++){
7             if(i==1 || i==n || j==1 || j==n ){
8                 printf("*");
9             }
10            else{
11                printf(" ");
12            }
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

QUESTIONS 18 / 18 completed

Hollow Diamond ▾

HOLLOW
DIAMOND

Write a program that prints a hollow diamond pattern of stars using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

INPUT

8

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     for(int i=1;i<=n;i++){
6         for(int j=1;j<=n-i;j++){
7             printf(" ");
8         }
9         for(int k=1;k<=(2*i-1);k++){
10             if(k==1 || k==(2*i-1)){
11                 printf("*");
12             }
13             else{
14                 printf(" ");
15             }
16         }
17         printf("\n");
18     }
19     for(int i=n-1;i>1;i--){
20         for(int j=1;j<=n-i;j++){
```

```
21         printf(" ");
22     }
23     for(int k=1;k<=(2*i-1);k++){
24         if(k==1 || k==(2*i-1)){
25             printf("*");
26         }
27         else{
28             printf(" ");
29         }
30     }
31     printf("\n");
32 }
33 return 0;
34 }
```

QUESTIONS 18 / 18 completed

Pascal's triangle ▾

PASCAL'S
TRIANGLE

Write a program that prints a Pascal's triangle pattern using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

INPUT

10

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     for(int i=0;i<n;i++){
6         for(int j=0;j<n-i-1;j++){
7             printf(" ");
8         }
9         int val=1;
10        for (int j=0;j<=i;j++){
11            printf("%d ",val);
12            val=val*(i-j)/(j+1);
13        }
14        printf("\n");
15    }
16    return 0;
17 }
```

QUESTIONS 18 / 18 completed

zigzag ▾

ZIGZAG

Write a program that prints a zigzag pattern of stars using nested loops.

PROGRAM EDITOR

C ▾

Run

Save

INPUT

8

```
1 #include <stdio.h>
2 int main(){
3     int n,i,j;
4     printf("Enter the number of columns: \n");
5     if(scanf("%d",&n)!=1){
6         return 0;
7     }
8     for (i=1;i<=3;i++){
9         for(j=1;j<=n;j++){
10            if(((i+j)%4==0) || (i==2 && j%4==0)){
11                printf("* ");
12            }
13            else{
14                printf(" ");
15            }
16        }
17        printf("\n");
18    }
19    return 0;
20 }
```

QUESTIONS 18 / 18 completed

Spiral pattern

SPIRAL PATTERN

Write a program that prints a spiral pattern of numbers using nested loops.

PROGRAM EDITOR

C

Run

Save

INPUT

4

```
21     for(i=bottom;i>=top&&left<=right;i--)
22         a[i][left]=num++;
23     left++;
24 }
25 printf("\nSpiral Pattern:\n");
26 for(i=0;i<n;i++){
27     for(j=0;j<n;j++){
28         printf("%4d",a[i][j]);
29     }
30     printf("\n");
31 }
32 return 0;
33 }
```

QUESTIONS 18 / 18 completed

Armstrong

ARMSTRONG

Write a program to find the given number is Armstrong or Not.

PROGRAM EDITOR

C

Run

Save

INPUT

153

```
1 #include <stdio.h>
2 int main(){
3     int num,originalNum,result=0,remainder;
4     scanf("%d",&num);
5     originalNum=num;
6     while(originalNum!=0){
7         remainder = originalNum%10;
8         result += remainder*remainder*remainder;
9         originalNum/=10;
10    }
11    if(result==num){
12        printf("Armstrong\n");
13    }
14    else{
15        printf("Not Armstrong\n");
16    }
17    return 0;
18 }
```

QUESTIONS 18 / 18 completed

Perfect

PERFECT

Write a program to find
the given number is
Perfect number or Not.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 int main(){
3     int num,sum=0;
4     scanf("%d",&num);
5     for(int i=1;i<=num/2;i++){
6         if(num%i==0){
7             sum+=i;
8         }
9     }
10    if(sum==num && num>0){
11        printf("Perfect number\n");
12    }
13    else{
14        printf("Not a Perfect numnber\n");
15    }
16    return 0;
17 }
```

INPUT

6